



Areas covered	Programming Fundamentals Basics
Practical Number	07
Module	CS102.3-Programming Fundamentals – 26.1

Question 1 – Student Performance Analyzer

Write a Python program that prompts the user to enter marks for **6 subjects**. The program should calculate the total and average marks. Additionally, count how many subjects have marks below 50. If the average is greater than or equal to 60 and no subject has failed, display “Excellent Performance”. If the average is greater than or equal to 50, display “Pass”. Otherwise, display “Fail”.

Question 2 – ATM System with Limited Attempts and Balance Check

Write a Python program that simulates an ATM system. The program should allow a user to enter a PIN with a maximum of **3 attempts**. The correct PIN is “1234”. If the PIN is correct, allow the user to enter a withdrawal amount and check the following conditions:

- Withdrawal amount must not exceed balance
 - Minimum remaining balance should be Rs. 500
- Display appropriate messages for each case.

Question 3 – Supermarket Billing System

Write a Python program that allows the user to enter the price and quantity of **5 items**. Calculate the total bill. Apply discounts as follows:

- Total above Rs. 10000 → 25%
 - Rs. 5000 – 10000 → 15%
 - Below Rs. 5000 → No discount
- Display total, discount, and final amount.

Question 4 – Bank Transaction Monitor

Write a Python program that continuously accepts deposit or withdrawal amounts. Positive values represent deposits and negative values represent withdrawals. The program should stop when the user enters 0. Display the final balance and indicate whether the account is “Overdrawn” (balance < 0) or “Active”.

Question 5 – Electricity Bill Calculator with Slab Rates

Write a Python program to input electricity units for **4 months** and calculate total usage. Apply billing:

- First 100 units → Rs. 10/unit
 - Next 100 units → Rs. 15/unit
 - Above 200 units → Rs. 20/unit
- Display total bill and usage category (High if >400 units).

Question 6 – Inventory Management System

Write a Python program that allows the user to enter stock quantities for **6 products**. Count:

- Products with stock < 10 (Low stock)
- Products with stock between 10–50 (Normal)
- Products with stock > 50 (High stock)

Question 7 – Temperature Alert System

Write a Python program that continuously accepts temperature readings until -1 is entered. Display:

- Number of high temperatures (>35)
- Number of normal temperatures (20–35)
- Number of low temperatures (<20)

Question 8 – Employee Salary and Bonus System

Write a Python program to input salary for **5 employees**. Calculate bonus:

- Salary $\geq 100000 \rightarrow 20\%$
 - Salary $\geq 50000 \rightarrow 10\%$
 - Otherwise $\rightarrow 5\%$
- Display total salary payout and total bonus given.